Saverio Mattia **Merenda**

AI Solution Architect · M.S. in Computer Science

Reggio Emilia - Parma, Italy

info@merendamattia.com | merendamattia.com | merendamattia | merendamattia | Saverio Mattia Merenda

Education

University of Parma
Parma, Italy

M.S. in Computer Science, Grade: 110/110 *cum laude* — *Sep. 2024 - Jul. 2026*

- Thesis Title: *Static Detection of Access Control Incompleteness in EVM Smart Contracts: Relational Taint Analysis for Cross-Chain Bridges*.
- Best Marks: **Software Verification**, **Artificial Intelligence**, **Compilers**, **Quantum Computing**, **Big Data**.
- Awarded **two merit-based ER.GO scholarships**, granted on the basis of academic excellence and income criteria.

University of Parma
Parma, Italy

B.S. in Computer Science, Grade: 108/110 — *Sep. 2021 - Jul. 2024*

- Thesis Title: *Construction of sound Control-Flow Graphs for bytecode EVM*.
- Best Marks: **Calculus**, **Programming**, **Algorithms & DS**, **Cloud Administration**, **Database**, **AI**, **HPC**.
- Awarded **three merit-based ER.GO scholarships**, granted on the basis of academic excellence and income criteria.

Pubblications

2025 — **Blockchain: Research and Applications**, V. Arceri, S. M. Merenda, L. Negrini, L. Olivieri, E. Zaffanella: *"EVMLiSA: Sound Static Control-Flow Graph Construction for EVM Bytecode".* (doi: 10.1016/j.bcra.2025.100384) — *Journal paper*

2024 — **FTfJP@ISSTA/ECOOP 2024**, V. Arceri, S. M. Merenda, G. Dolcetti, L. Negrini, L. Olivieri, E. Zaffanella: *"Towards a Sound Construction of EVM Bytecode Control-flow Graphs".* (doi: 10.1145/3678721.3686227) — *Workshop paper*

Experience

Keplero AI
Milan, Italy

AI Solution Architect — *From Nov. 2025*

- Design, development, and deployment of **custom AI agents and AI solutions** for client-specific use cases and internal business process optimization.
- Creation of **autonomous AI agent ecosystems**, including meta-agents capable of automatically configuring and deploying AI agents for customers (agents that create agents).
- Development of **continuous learning AI agents for customer support**, enabling adaptive behavior through automated knowledge updating and learning pipelines.
- Engineering of **ad-hoc MCP tools and internal AI tooling platforms** to support scalable agent configuration, orchestration, and lifecycle management.
- Integration of AI agents into operational workflows using modern orchestration and automation platforms (e.g., n8n), enabling scalable and production-ready deployments.

University of Parma
Parma, Italy

Researcher and Software Developer (Master's Degree internship) — *Jan. 2025 - May 2026*

- Extension of **EVMLiSA** to support cross-chain analysis, enabling vulnerability detection across multiple blockchain ecosystems.
- Design and implementation of a static checker for **Access Control vulnerabilities**, with a specific focus on **Access Control Incompleteness** in cross-chain bridge smart contracts, based on **relational taint analysis**.
- Evaluated the checker on **16 cross-chain bridges** (1,003 contracts), achieving **100% recall** and precision ranging from 73.24% (ABI-only) to **94.36%** in the modifier-aware configuration.

University of Parma
Parma, Italy

Teaching Assistant — *Sep. 2024 - Jan. 2026*

- Teaching Assistant for the "Fondamenti di Programmazione A + B (15 CFU)" bachelor course in the Computer Science degree.
- Delivered 45+ hours of frontal lectures.

University of Parma
Parma, Italy

Researcher and Software Developer (Bachelor's Degree internship) — *Sep. 2023 - May 2025*

- Contributed to the **development of EVMLiSA**, a static analyzer for Ethereum smart contracts built on the LiSA (Library for Static Analysis) framework, using abstract interpretation to construct sound Control-Flow Graphs (CFGs).
- **Designed and implemented** multiple security checkers to detect common vulnerabilities such as reentrancy, randomness dependency, enhancing the analyzer's precision and robustness.
- Applied **advanced design patterns** and **software architecture principles** to improve the scalability and maintainability of the tool.
- The project led to two peer-reviewed publications: the workshop paper *FTfJP@ISSTA/ECOOP (2024)* and the journal article *Blockchain: Research and Applications (2025)*.

HeliopsyLab *Reggio Emilia, Italy*

SOFTWARE DEVELOPER *Jan. 2021 - Sep. 2021*

- Developed and implemented cutting-edge management software for hospital facilities in the Lazio and Lombardia regions, streamlining operations and reducing administrative inefficiencies by 15%.
- Advanced new features within the Picasso and Mirth integration software, enhancing its capabilities and functionality.

Talks

INTERNATIONAL CONFERANCES AND WORKSHOP

2024 **Towards a Sound Construction of EVM Bytecode Control-flow Graphs**, 26th International Workshop on Formal Techniques for Java-like Programs, FTfJP 2024. [slides] *Vienna, Austria*

Research Projects

Research Participant, "LLMs Meet Static Analysis: improving quality and reliability of AI-generated code" *Parma, Italy*

ISCRA PROJECT (CLASS C), CINECA *2024*

The goal of the project is to conduct an extensive quality and safety evaluation of the code generated with some of the most popular and open-source LLMs employing static analyzers, that can detect vulnerabilities and run-time errors statically, without executing the code. Once this information is available, it will be included in the code-generation task, to guide the LLM itself to produce a more precise and safe output, in which static analysis is somehow introduced in the pipeline of the code-generation task.

Tools and Software

EVMLiSA: an abstract interpretation-based static analyzer for EVM bytecode

JAVA, GRADLE *From Sep. 2023*

EVMLiSA is a specific implementation of a static analyzer using the LiSA (Library for Static Analysis) library to conduct static analysis of Ethereum Virtual Machine (EVM) bytecode. In particular, it reconstructs a sound and precise Control-Flow Graph (CFG) directly from bytecode, resolving dynamic jump destinations (orphan jumps) via abstract interpretation. It ships security checkers for **reentrancy**, **timestamp dependency**, **tx.origin** vulnerabilities and **Access Control Incompleteness** in cross-chain bridges, and analyzes raw bytecode, local files or on-chain addresses (Etherscan API) as a CLI application, Docker container or Java library. EVMLiSA's primary objective is to provide semantic information and valuable warnings for developers and security auditors. EVMLiSA is distributed under the MIT license, and it is available on GitHub: `github.com/lisa-analyzer/evm-lisa`.

Events

2025 **Conference participant**, CSV 2025, 4rd Challenges of Software Verification Symposium 2025. [link] *Venice, Italy*

2024 **Student**, Lipari Summer School on Abstract Interpretation. [link] *Lipari, Italy*

2024 **Conference participant**, CSV 2024, 3rd Challenges of Software Verification Symposium 2024. [link] *Venice, Italy*

Academic Activity

Personal Financial AI Agent

PYTHON, OLLAMA, OPENAI, GOOGLE GEMINI *2025*

- Developed a system of interacting **AI agents** acting as a **personal financial advisor**, capable of providing multilingual financial guidance and portfolio analysis.
- Integrated multiple **LLM providers** (Ollama, Google Gemini, OpenAI) for flexible online or fully offline operation.
- Implemented **interactive conversations**, **financial profile extraction**, and **portfolio generation** through a web interface.
- Gained experience in **LLM orchestration**, **Python backend design**, and the integration of conversational AI with financial data analysis.
- Source Code: `github.com/merendamattia/personal-financial-ai-agent`

Isolette: Interactive Scenario Visualization and UI Simulation

C, VUE.JS, TYPESCRIPT *2025*

- Contributed to the **Isolette** project, a safety-critical system simulating a neonatal incubator that monitors and controls temperature.
- Designed and implemented the **User Interface Hardware** (UIH) and its simulator (UIHS) to ensure behavioral consistency during testing.
- Developed an **interactive web interface** using Vue and TypeScript with real-time temperature charts, timeline navigation, and JSON-based validation.
- Built a **Scenario Visualization Framework** replacing gnuplot, using Vue and Chart.js for dynamic, browser-based data visualization.
- The result is **a scalable, browser-based visualization frontend** that improves usability, speeds up development, and removes external dependencies.

Impact of Data Preprocessing on Neural Network Performance

Python — *2025*

- Investigated how **data preprocessing techniques**—including NaN handling, outlier removal, normalization, and feature transformation—affect the accuracy and stability of **feed-forward neural networks**.
- Implemented multiple preprocessing pipelines and benchmarked them on diverse **classification and regression** datasets.
- Gained strong experience in **data quality analysis**, **model evaluation**, and **experimental methodology**.
- Source Code: `github.com/merendamattia/neural-network-performance-by-data-quality`

GraphQL API Design and Comparative Analysis

GraphQL, REST — *2025*

- Designed and implemented a **GraphQL API**, defining schema, queries, and mutations for flexible and efficient data retrieval.
- Compared **GraphQL** and **REST** architectures, analyzing efficiency in terms of data transfer, query complexity, and latency.
- Highlighted GraphQL's benefits in reducing over-fetching and under-fetching, improving client-side data control.
- Strengthened skills in **API design**, **backend integration**, and **data modeling**.

Quantum Portfolio Optimization

Python — *2025*

- Explored **quantum algorithms** for portfolio optimization using **QAOA** and **VQE**.
- Formulated the optimization problem as a **Quadratic Unconstrained Binary Optimization (QUBO)** model and implemented simulations using **Qiskit**.
- Compared performance under ideal and noisy conditions, discussing the potential of **quantum computing in finance** and its current hardware limitations.
- Source Code: `github.com/merendamattia/quantum-portfolio-optimization`

Optimization of Academic Guarantors: a Declarative Approach

ASP, Python — *2024*

- Developed an **automated system** for assigning academic guarantors to university courses, ensuring compliance with ministerial regulations.
- Designed a **data preprocessing pipeline** and encoded the optimization logic in **Answer Set Programming (ASP)** to balance resource allocation.
- Demonstrated the effectiveness of **declarative programming** in solving complex institutional optimization problems.
- Source Code: `github.com/merendamattia/ottimizzazione-garanti-accademici`

Deep Neural Network

C++ — *2024*

- Developed a **neural network framework** supporting customizable architectures, activation functions, and training via **backpropagation** and **gradient descent**.
- Implemented **model serialization** and loading to facilitate retraining and experiment reproducibility.
- Gained solid experience with **machine learning fundamentals**, **optimization algorithms**, and **C++ object-oriented design**.
- Source Code: `github.com/merendamattia/deep-neural-network`

Extracurricular Activity

My-gpt4

Python — *2023*

- Crafted a cutting-edge Python integration leveraging multiple reverse-engineered language-model APIs, contributing to the decentralization of the AI industry.
- Source Code: `github.com/merendamattia/my-gpt4`

Tracking Messages on Bitcoin Blockchain using OP-RETURN field

JS, HTML — *2022*

- Devised an innovative algorithm to securely trace immutably written messages within the Bitcoin blockchain, utilizing transaction hashes for enhanced transparency.
- Demo: `merendamattia.com/dev/btc/tracking/`
- Source Code: `github.com/merendamattia/op-return-tracking-message-bitcoin`

Development of Various Bots

Python, JS — *From 2021*

- Developed a sophisticated bot designed to navigate financial markets, employing strategic approaches to optimize profits while minimizing potential losses.
- Customized Telegram bots designed for personal optimization, streamlining and enhancing my daily routine.